

NAVEEN MUTHUKUMAR

Full-Stack Developer | React.js | Next.js | FastAPI | REST API | RabbitMQ | Docker | CI/CD
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PROFESSIONAL SUMMARY

Results-driven Full-Stack Developer with 2+ years of production-grade experience building scalable web applications using React.js, Next.js, FastAPI, and Python. Proven track record of independently owning full modules end-to-end across LMS and SIS platforms, delivering measurable outcomes such as a 70% reduction in manual content-creation time and a 4-role RBAC architecture later adopted as a system-wide standard. Skilled in REST API design, WebSocket real-time systems, RabbitMQ messaging, AI-powered feature development (Claude AI / Claude Code), Firebase authentication, Docker, CI/CD, and Agile/Scrum collaboration.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript (ES6+), Python, React.js, Next.js, FastAPI, Tailwind CSS, SQL

APIs & Real-Time: RESTful API Design, WebSockets, JSON, Async/Await, Axios

Messaging & DevOps: RabbitMQ, Docker, CI/CD Pipelines, Pre-commit Hooks, Git/GitHub, Version Control

AI & Platform Engineering: Claude AI Integration, Claude Code (AI Dev Agent), LMS/SIS Platform Dev, Firebase Authentication

Databases & Practices: MySQL, PostgreSQL, Debugging, Unit Testing, Code Review, SOLID Principles

Methodologies: Agile, Scrum, SDLC, Feature-Branch Workflow, RBAC & Workflow Design, Cross-Browser Testing

WORK EXPERIENCE

Full-Stack Developer (Frontend Focus) | *iNoesis Technologies Private Limited — Chennai* | **Aug 2023 – Present**

Project 1: eCampusBuddy — Student Information System (SIS), Document Management Module

- Independently owned and engineered a document management module end-to-end using Next.js and FastAPI with Claude Code as an AI development agent, applying domain expertise in admissions and document-verification workflows.
- Redesigned the document workflow to fix mismatched status vocabularies, a non-functional verification step, and missing role-based access controls.
- Designed a 6-stage, dual-axis document status model (Not Uploaded, Uploaded, Under Review, Verified, Rejected, Resubmit Required) and a role-based capability matrix spanning 4 roles (Student, Staff, Coordinator, Admin) — later adopted as a system-wide architectural standard across other SIS modules.
- Corrected progress-tracking logic to count verified, not merely uploaded, documents, improving case-readiness accuracy for staff and coordinators.
- Implemented RabbitMQ for asynchronous document processing, containerized the module with Docker, and set up CI/CD fundamentals and pre-commit hooks for code-quality gates.

Project 2: Digital Twin — AI-Powered Learning Management System [Current]

- Built a full-featured LMS using Next.js and FastAPI, supporting course management, enrollment, progress tracking, and instructor dashboards.
- Integrated Claude AI for an intelligent assignment-generation module, auto-creating contextual assignments from course materials and reducing instructor workload.
- Implemented Firebase Authentication with custom claims and RBAC across 3 user roles (Students, Instructors, Admins), adding session persistence and protected route middleware for end-to-end access control.
- Built RESTful FastAPI endpoints for course lifecycle, submissions, and grading using async Python; applied component-based Next.js architecture with SSR/SSG for performance and SEO.
- Delivered an alumni engagement module for networking and mentorship matching within the LMS ecosystem.

Project 3: quicktest.diy — AI-Powered Quiz Generation Platform

- Architected an AI-powered quiz platform (React.js, Next.js, Tailwind CSS) with automated question generation from documents, cutting manual content-creation time by approximately 70%.
- Built a WebSocket-based real-time layer for live quiz sessions with instant score updates, and integrated RESTful APIs via Axios for authentication, quiz lifecycle, and question-bank operations.
- Designed a reusable, modular UI component library following SOLID principles; added RBAC, student invitation workflows, and an analytics dashboard for educator performance tracking.
- Applied lazy loading, code splitting, and React memoization across the platform, improving page load speed and Core Web Vitals scores.

EDUCATION

Bachelor of Computer Applications (BCA) — 2024–2027 (In Progress), Madras University – Distance Education, Chennai

Diploma in Electrical Engineering — 2019–2022, Alagappa Polytechnic College

CERTIFICATIONS

Full Stack Developer – MERN Stack Certification | Besant Technologies, Chennai